

Technology Stack

Date	4 July 2026
Team ID	XXXXXX
Project Name	Human Development Index (HDI) Prediction Using Machine Learning
Maximum Marks	2 Marks

Technology Stack Details:

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML, CSS	Used to design a simple, responsive, and user-friendly interface where users can enter HDI input values and view prediction results.

2	Backend / Server-Side	<i>Python, Flask</i>	<i>Python is used to build the machine learning model, and Flask connects the frontend with the trained model to process user requests.</i>
3	Database / Data Storage	<i>CSV Dataset (hdi_dataset.csv) and Pickle Model (model.pkl)</i>	<i>The CSV file stores the training data, while the Pickle file stores the trained machine learning model for prediction.</i>
4	Cloud / Hosting / Deployment	<i>Flask Local Server (or Render/PythonAnywhere if you deployed it)</i>	<i>Used to host and run the web application so users can access the HDI prediction system.</i>
5	Version Control & CI/CD	<i>GitHub</i>	<i>Used to store the project source code, track changes, and collaborate with team members.</i>

6	Third-Party APIs / Other Tools	<i>Scikit-learn, Pandas, NumPy, Visual Studio Code</i>	<i>Scikit-learn is used for Linear Regression, Pandas and NumPy handle data processing, and Visual Studio Code is used for development.</i>
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